

Aleksandra Winkler

aleksandra.winkler@utah.edu

aleksandrawinkler.github.io/

Education

- 2024-2027 **University of Utah, B.Sc. Physics**
Overall GPA: 3.912; **Major GPA:** 3.916
Coursework: Cosmology, Observational Astronomy, Foundations Astronomy, Relativity, Kinematics, Classical Mechanics, Electrostatics, Quantum Mechanics, Thermodynamics, Differential Equations, Linear Algebra, Complex Variables, PDE's
- 2023-2024 **Weber State University**
Overall GPA: 3.88
Coursework: Calculus Series (I, II, III), Introductory Astronomy, College Algebra, Trigonometry, Kinematics, Mechanics

Awards & Honors

- 2023-2026 **Dean's List, 6 Semesters**
- 2023-2024 **Utah NASA Space Grant Consortium awardee:** Awarded to outstanding scholars pursuing a STEM degree
STEM Underrepresented Scholarship: Awarded to scholars who are included in groups that are underrepresented in STEM
Doris Parker Women in Physics Scholarship: Awarded scholarship for being an outstanding female scholar pursuing a physics degree

Research Experience

- 2025-
Present Undergraduate Research Assistant, University of Utah, (Mentors: [Dr. Catherine Manea](#) and [Dr. Gail Zasowski](#))

- Determined high precision (uncertainties $<1\%$) abundances of stars in 12 wide binaries using strictly line-by-line differential analysis of optical high resolution spectra with BACCHUS
- Identified two wide binary systems with anomalous (nearly an order of magnitude) abundance difference in lithium
- Performed spectral energy distribution modeling to search for color excess to search for hidden tertiary companions
- Used orbital modeling code "The Joker" to fit radial velocity time-series data to search for hidden tertiary companions

2025 Undergraduate Research Assistant, University of Utah, (Advisor: [Dr. Kurt Vinhage](#))

- Conducted research on what types of matrices are included in the Lorentzian Group in collaboration with 3 other undergraduates
- Conceptualized how the tilt of the light cone can be used to describe the effect black holes have on space-time through SageMath coding simulations
- Applied extensive linear algebra, discrete math, and differential geometry concepts to classify Lorentz transformations and showed how this could be interpreted into a physical representation of a light cone

Observing Experience & Proposals

Accepted Large Public Facility Proposals:

- 2026 **McDonald Observatory**, Harlan J. Telescope, Tull Coude Spectrograph
Unveiling the Origin and Precise Nature of the Li Dip Using Wide Binaries
PI- C. Manea. 60 hours allocated, 6 observing nights.
- 2026 **Gemini Observatory**, Gemini North Telescope, MAROON-X Spectrograph
Hunting for a Compact Companion in a Lithium-Rich Potential Triple Stellar System
PI- C. Manea. 2.3 hours allocated.
- 2026 **Cerro Tololo Inter-American Observatory**, SOAR Telescope, STELES Spectrograph
Distinguishing Between Li-Enrichment Scenarios in a Remarkable (Potential) Triple System with STELES
PI- C. Manea. 4.7 hours allocated.

Experience:

- 2026 **6 nights (~60 hours) of observing** with Tull Coude Spectrograph on 2.7m telescope at the McDonald Observatory

Other Experience

- 2025 **Physics Lab Learning Assistant** University of Utah, Salt Lake City, Utah
- Helped peers discover core physics topics during their laboratory time and answer questions to my fullest ability
 - Clarified confusion on MatLab code for peers who are brand new to the language and concepts
 - Provided an encouraging space for fellow undergrad students to ask questions and feel very open to learning
- 2023-2025 **Ballet Instructor** The LAB Dance Studio, Layton, Utah
- Instructed hundreds of students ages 5-18 on core classical ballet movement principals
 - Encouraged group work and leadership to enhance confidence within young individuals
 - Choreographed dances for the students to preform and compete, learned how to instruct individuals in a high pressure environment
- 2022-2023 **Professional Ballet Dancer** Ballet Metropolitan, Columbus, Ohio
- Collaborated with dancers, directors, and repetiteurs to produce ballet performances multiple times a year
 - Self corrected mistakes in technique and style to improve overall as a dancer, learned to be very detail oriented
 - Familiarity with being involved in a high stakes environment and determination towards achieving a major goal

Tools & Software

- **Programming & Tools:**
Python (NumPy, SciPy, Matplotlib, Astropy), MATLAB, LaTeX, Linux, Git/GitHub, The Joker, starfuSED
- **Data & Analysis:**
Stellar Spectroscopy, Differential Abundance Analysis, Atmospheric Parameter Generation, FITS Data Processing, Spectral Energy Distribution Analysis, Telescope Operation, Brussels Automatic Code for Characterizing High-accuracy Spectra (BACCHUS), Astronomical Image Processing

Last updated: September 23, 2026